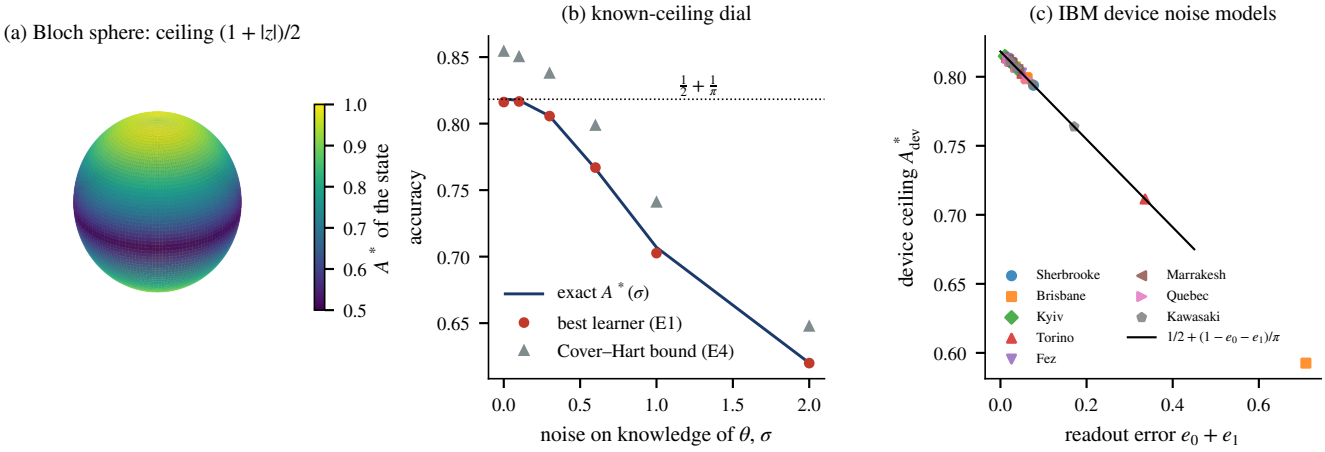

(a) Bloch sphere: ceiling $(1 + |z|)/2$
1.0
0.9
0.8
0.7
0.6
0.5
A^* of the state
(b) known-ceiling dial
accuracy
0.85
0.80
0.75
0.70
0.65
$\frac{1}{2} + \frac{1}{\pi}$
exact $A^*(\sigma)$
best learner (E1)
Cover–Hart bound (E4)
0.0
0.5
1.0
1.5
2.0
noise on knowledge of θ, σ
(c) IBM device noise models
device ceiling A^*_{dev}
0.80
0.75
0.70
0.65
0.60
Sherbrooke
Brisbane
Kyiv
Torino
Fez
Marrakesh
Quebec
Kawasaki
$1/2 + (1 - e_0 - e_1)/\pi$
0.0
0.2
0.4
0.6
readout error $e_0 + e_1$